Darian Abreu, Engineer-in-Training

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Mechanical Engineer with 7+ years of experience specializing in CAD automation, parametric process equipment design, and manufacturing optimization, with expertise in Autodesk Inventor, AutoCAD and Excel VBA.

EDUCATION

University of Delaware, Newark, DE May 2017
College of Engineering
Bachelor of Science
Major: Mechanical Engineering | Minors: Electrical and Computer Engineering, Mathematics
Cumulative GPA: 3.38 **Major GPA**: 3.35
Achievements or Awards: 1st Place Design (ADVIA 1800), Environmental Design Discipline Award (Undercarriage Decontamination)

RELEVANT COURSE PROJECTS

Undercarriage Decontamination System, United States Department of Agriculture August 2016 - May 2017

- Made an easy to build, cost-effective, open-source undercarriage decontamination system to disinfect vehicles and prevent disease outbreaks on small farms in the United States by diminishing viral load of a disease by at least 90%
- Re-engineered comparable market designs to cut down cost by over 99% while utilizing only 6% of provided budget

Beach Wheelchair, Dr. Peter Popper February 2016 - May 2016

- Produced an affordable, transportable solution for wheelchair users to maneuver on sand and disassemble into approximately 13 cubic feet of storage volume
- Collaborated with a 6-engineer team to devise a beach wheelchair concept at an 80% lower price-point than market benchmarks, minimizing parts and using material analyses to optimize two-wheel design

ADVIA 1800 Impeller Drive Mechanism, Siemens Healthcare Diagnostics May 2016 - February 2016

- Modernized the Siemens ADVIA 1800 drive mechanism to allow adjustable horizontal and vertical impeller positioning, while meeting strict accuracy benchmarks for rapid chemical analysis of distinct fluids
- Implemented spring-loaded mechanism with an injection-molded cam to implement a cost-effective solution to previously, fixed single-plane motion impeller handling up to 1800 tests per hour

EXPERIENCE

Koch Modular Process Systems Paramus, New Jersey
CAD Designer June 2025 - Present

- Translating process markups into 200+ client-ready P&IDs and PFDs across 24+ projects with same-day AutoCAD turnaround
- Compiling AutoCAD dynamic blocks, LISP routines, and Excel macros to automate repetitive drafting tasks and streamline P&ID and PFD workflows, decreasing production time
- Drafting power, instrument loop, control system and electrical overview drawings for engineering design deliverables in support of the EIC department

Mechanical Engineer December 2024 - June 2025

- Developed constrained parametric 3D models of Karr & Scheibel column internals in Autodesk Inventor, reducing design iteration time across multiple extraction configurations
- Created detailed 3D assemblies and fabrication drawings for Karr and Scheibel column internals including reciprocating plates, agitators, shaft assemblies, packing sections and liquid distribution components using Autodesk inventor
- Recreated legacy AutoCAD drawings in Autodesk Inventor with near-identical accuracy to original production standards

Skyline Windows Bronx, New York
Volume Project Engineer July 2024 - December 2024

- Consulted with architects to tailor window and door solutions ensuring optimal fit, functionality and aesthetic appeal
- Assessed building envelope, glass deflections and structural integrity to mitigate wind loads and devise a project scope
- Analyzed field measurements to optimize sizing, minimize manufacturing complexity and improve overall project efficiency
- Planned regular site visits to ensure accurate installation and address any potential challenges promptly

Custom Production Supervisor March 2022 - July 2024

- Automated air and water infiltration testing, eliminated error-prone handwritten records by developing a VBA-assisted spreadsheet adhering to ASTM standards for air leakage testing (E547) and water infiltration testing (E283, E331)
- Conducted thorough shop drawing reviews to ensure design feasibility and generate comprehensive factory order packages
- Processed 1000+ façades, including renowned landmarks such as Giorgio Armani's Penthouse, the Hampshire House & NYU
- Implemented dynamic AutoCAD details to efficiently construct elevation drawings for interior, exterior and internal muntins
- Wrote extensive Visual Basic .NET scripts to autonomously create proper window hardware and field installation materials
- Oversaw a 1000% increase in custom department's gross profit first year, producing nearly $20 million in revenue

Manufacturing/Estimating Engineer July 2021 - March 2022

- Provided material takeoffs, financial forecasts, estimations, processing documentation and labor expenses for production
- Organized OneNote manual for engineering team to collaborate and standardize order entry with real-time updates
- Utilized macros for glass pricing, inventory checks and audits, minimizing manual errors and improving operational efficiency

Public Consulting Group Remote
Contact Tracer December 2020 - July 2021

- Designed spreadsheet for team to input, monitor and sort 2400 contacts by quarantine date using Excel
- Managed biweekly data entry of school exposure line lists of 80+ students and staff in liaison with specialist and supervisor
- Developed technical documentation, managed workflows, and provided training and support to more than 20 employees

Frank's Drywall Rockland County, New York
Construction Manager June 2019 - December 2020

- Streamlined 3+ process improvements during first year in position, including standardizing customer service model
- Built functions for Google Sheet to dynamically generate invoices/estimates and autonomously produce customer-email interaction with PDF attachment
- Interpreted plans, studying 20+ isometric drawings to formulate estimates for gypsum board, acoustical ceilings, or framing

Aquatectonic/Lothrop Associates LLP White Plains, New York
Design/Technical Engineer November 2017 - June 2019

- Coordinated and interpreted multi-disciplinary drawings to construct sections, plans, schedules, and P&ID diagrams for diverse pools, spas, cold plunges, water features, steam rooms and saunas, including 1st zero-entry steel wading-pool
- Defined and detailed fittings, lights, main drains, gutters, skimmers, surge tanks, ladders, speakers, entryways, seating construction and all relevant equipment to meet codes and approval standards reviewed by the NYS Dept. of Health
- Determined equipment specifications, sizing, and system filtration designs while performing hydraulic calculations to specify critical pumps, pipes, and utility requirements

ACHIEVEMENTS & VOLUNTEER INVOLVEMENT

Engineer-in-Training, State of New York, Certification #1716704 June 2018 - Present
Dean's List Fall 2013, Spring 2014, Fall 2015, Spring 2016, Fall 2016, Spring 2017
Benjamin E. Herring Scholarship May 2016
University of Delaware Presidential Scholarship May 2013
American Society of Mechanical Engineers, Member September 2015 - Present
American Cancer Society, Volunteer September 2014 - Present

SKILLS

- Computer & Software: Adobe Acrobat, Amazon Connect, Arduino, AutoCAD, AutoDesk Inventor, Beamchek, Bluebeam Revu, Cambridge Engineering Selector, CommCare, Excel VBA, GitHub, HTML, JavaScript, Kofax Power PDF, LabVIEW, Linux, MATLAB, Microsoft Office, Microsoft Teams, PLC Programming, PowerBI, PrefSuite: (CAD, Gest, Wise), Python, Revit, SolidWorks (FEA), SQL, Visual Basic .NET
- Languages: Bilingual in Spanish (Reading, Writing and Speaking), Excellent Verbal, Written and Analytical Skills
- Machining Tools: Milling Machines, CNC, Lathes, Band Saws, Table Saws, Drill Presses, Calipers